

Juan David Muñoz-Bolaños

Rechenauerstraße 42, Innsbruck, Austria (Tirol)

+43 676 9115145

juan.munoz@i-med.ac.at

TTControl S.r.l.

Hiring Team

Julius-Durst Straße 66

I-39042 Brixen / Bressanone, Italy

Subject: Application for Software Developer – Computer Vision and Robotics (m/f/d)

Dear Hiring Team,

As a physicist and computer vision engineer with hands-on C++/Python experience in real-time hardware control and FPGA-adjacent embedded systems, I am applying for the Software Developer role at TTControl in Brixen. My goal: make machines smarter through reliable, precisely engineered perception.

During my Ph.D. at the Medical University of Innsbruck, I built automated C++/Python pipelines for optical metrology and hardware integration, from design and specification through to testing and documentation. At IPHT Jena, I applied supervised ML to hyperspectral image data for automated classification. At INTECOL S.A.S., I deployed Halcon-based vision pipelines for high-speed industrial inspection, resolving critical production failures through custom illumination and optics design.

Originally from Colombia, I have been working in Innsbruck for the past four years as a researcher at the Medical University, with a certified B2 German level and full integration into DACH working culture. With my Ph.D. completing in May, I am available to join your team in Brixen shortly thereafter. I adapt quickly to new toolchains, including embedded Linux and ROS 2 environments, with the help of modern AI development tools.

Sincerely,

Juan David Muñoz-Bolaños

Optical and Computer Vision Engineer

Medical University of Innsbruck

Juan David Muñoz-Bolaños

Optical and Computer Vision Engineer

Rechenauerstraße 42, Innsbruck, Austria (Tirol)

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juan.munoz@i-med.ac.at | [LinkedIn Profile](#)



PROFESSIONAL SUMMARY

Computer Vision and Optical Engineer with a strong foundation in C++/Python software development and real-time hardware control. Proven track record deploying machine vision and AI/ML solutions (including JAX for GPU acceleration) in both research and industrial environments, with experience in Labview FPGA, hardware-integrated perception pipelines, and strong problem-solving skills.

TECHNICAL COMPETENCIES

C++/Python and Real-Time Systems

Extensive experience developing real-time control software in C++ and Python, interfacing with hardware and managing complex embedded system architectures. Fast learner on new toolchains with modern AI-assisted development.

FPGA and Embedded Systems

Experience with Labview FPGA and real-time control architectures in optical instrumentation. Motivated to deepen expertise in Yocto, Buildroot, and ROS 2 production environments.

Machine Vision and AI Integration

Hands-on deployment of vision pipelines using Halcon and OpenCV. Applied JAX for GPU-accelerated computing and supervised ML to hyperspectral data. Strong interest in integrating AI/CV models on embedded systems.

Problem Solving and System Integration

Driven by strong analytical and problem-solving skills. End-to-end R and D experience from specification and design through testing and qualification, coordinating multi-component systems.

PROFESSIONAL EXPERIENCE

Medical University of Innsbruck

Innsbruck, Austria

PhD Researcher – Computer Vision and Embedded Systems

Jan 2022 – Present

- **Real-Time Software Architecture:** Designed and implemented end-to-end C++/Python software for automated real-time control of complex optical hardware, owning the full lifecycle from requirements to deployment. Experience with FPGA-adjacent control loops and fast signal acquisition pipelines.
- **Hardware/Software Integration:** Integrated adaptive optical components, precision motion stages, and scientific cameras into a unified automated system, with rigorous calibration and testing protocols.
- **Perception Pipeline Development:** Built automated wavefront sensing and phase retrieval pipelines applying signal processing and numerical algorithms to high-fidelity image data.
- **Cross-functional Collaboration:** Coordinated with researchers and hardware vendors (Thorlabs, Edmund Optics) ensuring reliable delivery of integrated systems for peer-reviewed publications.

Leibniz Institute of Photonic Technology (IPHT)

Jena, Germany

R and D Researcher – AI and Photonics

Jun 2020 – Dec 2021

- **AI/ML Integration:** Applied supervised machine learning to hyperspectral image data for automated tissue classification, bridging algorithmic research with practical laboratory implementation.
- **Systems Prototyping:** Designed and built a dispersive fiber probe Raman imaging spectrometer; calibrated point-scanning setups for clinical applications.

INTECOL S.A.S.

Medellín, Colombia

Junior Machine Vision System Engineer

Mar 2018 – Mar 2019

- **Industrial Vision Deployment:** Designed and deployed Halcon-based image processing pipelines for high-speed manufacturing inspection, resolving critical production failures through custom illumination and optics.
- **System Architecture:** Managed selection of cameras, sensors, and illumination, leading the full implementation of optical inspection setups on active production lines.

EDUCATION

Medical University of Innsbruck
Ph.D. in Adaptive Optics & Microscopy

Austria
Jan 2022 – Present

Friedrich Schiller University Jena
M.Sc. in Photonics

Germany
2019 – 2021

University of Cauca
B.Sc. in Physics Engineering

Colombia
2012 – 2018

PUBLICATIONS & AWARDS

- **Awards and Selections:** Selected for ASML and ZEISS Summer Schools; awarded COLFUTURO Grant 2026 for best Colombian professionals.
- Muñoz-Bolaños, J. D., et al. Design of a dispersive 1064 nm fiber probe Raman imaging spectrometer. *Applied Sciences*, 14(11), 4726 (2024).
- Muñoz-Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. *ACS Photonics* 12(1), 176–184 (2025).
- Sohmen, M., Muñoz-Bolaños, J. D., et al. Optofluidic adaptive optics in multi-photon microscopy. *Biomedical Optics Express* 14(4), 1562–1578 (2023).

LANGUAGES

English (Fluent / Professional),
German (B2 - Working Proficiency),
Spanish (Native)

REFERENCES

Prof. Monika Ritsch-Marte
Head of Institute for Biomedical Physics
monika.ritsch-marte@i-med.ac.at

Medical University of Innsbruck

Dr. Christoph Krafft
Group leader: Spectroscopy Imaging
christoph.krafft@leibniz-ipht.de

Leibniz Institute of Photonic Technology

AGREEMENT FOR GUESTS

to carry out a practical training
with

Mr. **Juan David Munoz Bolanos** Date of birth: **1994-06-04**
temporarily living at **Emil-Wölk-Str. 7, 07747 Jena** (guest),

Student of the Friedrich-Schiller-University Jena, Abbe School of Photonics, gets in execution of his practical course in the research department Spectroscopy and Imaging, working group Raman and Infrared Histopathology (0104) from 2020-06-02 up to 2020-12-31 admission to the necessary rooms within Leibniz-IPHT.

The use of further devices, media and other infrastructure takes place in agreement with the Heads of the respective departments.

Mr. Juan David Munoz Bolanos has to observe the security regulations for working and other regulations valid in Institute as well as instructions of the laboratory and clean room responsible persons of the Leibniz-IPHT.

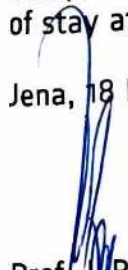
Mr. Juan David Munoz Bolanos has to treat all affairs and procedures strictly confidential. Copies, duplicates and others of business and technical information of each kind are forbidden, unless these are sanctioned by the responsible Department Heads.

Publications, reports and lectures about topics which belong together with the fields of work of Leibniz-IPHT require the previous written agreement of the Executive Board.

The liability of Mr. Juan David Munoz Bolanos is limited to cases of wilful action and gross negligence.

Mr. Juan David Munoz Bolaos is insured against accidents on the Friedrich-Schiller-University Jena, Abbe School of Photonics in the context of his/her practical course during his/her length of stay at Leibniz-IPHT.

Jena, 18 May 2020


Prof. Dr. Popp
Chairman
of Leibniz-IPHT


F. Sonderrmann
Executive Member
of Leibniz-IPHT


Juan David Munoz Bolanos
Guest

INTECOL

OPTIMIZING THE RHYTHM OF THE WORLD

Medellin, February 4, 2019

TO WHOM IT MAY CONCERN

Integración Tecnológica Industrial Intecol S.A.S. with Tax ID (NIT) 811.041.410-4 certifies that Mr. JUAN DAVID MUÑOZ BOLAÑOS, identified with Citizenship ID No. 1.061.772.278, has been employed by this company since July 3, 2018, under an indefinite term contract, serving in the position of Junior Project Engineer;

earning a salary of One million five hundred thousand pesos (\$1,500,000) plus three hundred thousand pesos (\$300,000) mobility allowance.

I will gladly address any additional information at the phone number 316 3322.

We appreciate your attention,

INTECOL

NIT. 811.041.410-4 - Tel.: 3163822

LENY ADRIANA RUIZ PINO

Administrative and Financial Director

Integración Tecnológica Industrial Intecol S.A.S.
www.intecol.com.co
Medellin Calle 42 #63c-82
PBX. (574) 316 3322 Cell: (57) 301 430 2532



URKUNDE

Die Friedrich-Schiller-Universität Jena verleiht durch die
Physikalisch-Astronomische Fakultät
mit dieser Urkunde

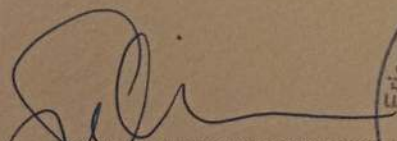
Herrn Juan David Munoz Bolanos
geb. 04.06.1994 in La Cruz (Kolumbien)
den Hochschulgrad

MASTER OF SCIENCE (M.Sc.)

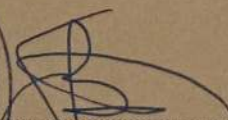
Er hat die Masterprüfung (120 ECTS)
im Studiengang »Photonics«

bestanden.

Jena, 20.12.2021


Prof. Dr. Christian Spielmann
Dekan




Prof. Dr. Silvana Botti
Vorsitzende des
Prüfungsausschusses

OFFICIAL TRANSLATION OF DOCUMENTS WRITTEN IN SPANISH, MADE 20
DECEMBER 2018 BY JUAN CARLOS OJEDA-GARCES, OFFICIAL TRANSLATOR
BY RESOLUTION No. 1171 ISSUED BY THE MINISTRY OF JUSTICE, REPUBLIC
OF COLOMBIA.



UNIVERSIDAD DEL CAUCA
ON BEHALF OF THE
REPUBLIC OF COLOMBIA
AN BY AUTHORIZATION OF THE MINISTRY OF NATIONAL EDUCATION
CONSIDERING THAT

JUAN DAVID MUÑOZ BOLAÑOS
CC 1.061.772.278 OF POPAYAN

HAS COMPLIED WITH ALL THE STATUTORY AND LEGAL REQUIREMENTS,
GRANTS THE TITLE OF

PHYSICAL ENGINEER

WITH ALL THE RIGHTS, PRIVILEGES AND DIGNITIES THAT AUTHORIZE HIM
FOR THE PROFESSIONAL EXERCISE

POPAYAN, 16 MARCH 2018

REGISTERED IN THE DIPLOMA BOOK No. 082 FOLIO 437 DIPLOMA No. 396
RESOLUTION No. R-266-07 MINUTE No. 14-07

(SIGNED)

THE PRINCIPAL OF THE UNIVERSITY
THE DEAN OF THE FACULTY
THE SECRETARY GENERAL OF THE UNIVERSITY

Juan Carlos Ojeda - Garcés
TRADUCTOR INTERPRETE OFICIAL
RESOLUCIÓN No 1171
MINISTERIO DE JUSTICIA 1997

Juan Munoz Bolanos

has participated in the

**ZEISS European
Summer School**

Lithography Optics &
Semiconductor Technologies

26 - 27 June 2025

ZEISS Semiconductor Manufacturing Technology, Oberkochen, Germany

Thomas Marschner

Dr. Thomas Marschner
Program Manager Funding Projects

Thomas Stammer

Dr. Thomas Stammer
CTO & Head of Product Strategy



Zertifikat

Muñoz-Bolaños JUAN DAVID

geboren am 04.06.1994

in La Cruz / Kolumbien

hat das Modul **Schriftliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden



Datum: 16.09.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA



ÖSD ist Mitglied von



ö s d

österreich schweiz deutschland

ID-Nummer: ZB2Ö25s34291

Zertifikat

Juan David MUÑOZ BALAÑOS

geboren am 04.06.1994

in La Cruz / Kolumbien

hat das Modul **Mündliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden



Datum: 11.08.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA



ÖSD ist Mitglied von



ösd

österreich schweiz deutschland



ID-Nummer: ZB2Ö25m40434

Passaporto № / Passport No.

MUNOZ BOLANOS

COLOMBIANA

Sexo / Sex
Lugar de nacimiento / Place of birth

Fecha de expedición / Date of issue

Fecha de vencimiento / Date of expiry

G. ANTIOQUIA

Dear Family

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